SOLUTIONS

Sample Question Paper-1

ECONOMICS

Section - A [16 Marks]

Answer 1

(i) **Option (c) is correct.**

Explanation: An indifference curve is convex to the origin because of diminishing Marginal Rate of Substitution. Marginal Rate of Substitution continuously decreases because of the law of diminishing marginal utility. The marginal rate of substitution is the rate at which a consumer gives up one good for another.

(ii) **Option (a) is correct.**

Explanation: Supply increases by 24 percent due to 8 percent increase in price means, price elasticity of supply is greater than unitary. Thus, supply curve will be positively sloped with positive intercept.

(iii) **Option (b) is correct.**

Explanation: The above curve represents the increase in demand due to rise in expected forthcoming price of the commodity. Increase in demand refers to a rise in the demand of a commodity caused due to any factor other than the own price of the commodity.

(iv) **Option (c) is correct.**

Explanation: A perfectly elastic price demand curve is parallel to the X axis. It means the quantity demanded changes without any change in the price of the commodity.

(v) **Option (d) is correct.**

Explanation: From the above options, Wealth Tax and Income Tax are direct taxes. A direct tax is a type of tax that a person or organization pays directly to the government and these are non-transferable taxes.

(vi) **Option (d) is correct.**

Explanation: From the above, no one should be included in national income. Transfer Earnings and Sale proceeds of Shares &Bonds is a financial claim and does not contribute to any productive activity. Black money includes all funds earned through illegal activity. That's why it is not included in the calculation of National Income.

(vii) **Option (d) is correct.**

Explanation: Government also intervenes in the process of price determination through Price Floor. Price floor refers to the minimum price (above the equilibrium price), fixed by the government, which the producers must be paid their produces.

(viii) **Option (d) is correct.**

Explanation: According to MR – MC approach, producer's equilibrium refers to stage of that output level at which MR = MC. A firm produces its profit maximising level of output only when MC = MR, and MC is greater than MR after MC = MR output level.

(ix)

Income (Y) ₹	Consumption (C) ₹	Saving ₹	MPC ₹	MPS ₹
8,000	6,800	1,200	–	–
10,000	8,000	2,000	$\frac{\Delta C}{\Delta Y} = \frac{1,200}{2,000} = 0.6$	$\frac{\Delta S}{\Delta Y} = \frac{800}{2,000} = 0.4$

Saving = Income – Consumption or S = Y – C

$MPC = \frac{\Delta C}{\Delta Y}$ $\quad$ $MPS = \frac{\Delta S}{\Delta Y}$

(x) Investment multiplier refers to the number of times by which the increases in output or income exceeds the increase in investment. It is measured as the ratio between change in output/income and change in investment.

$$K = \frac{\Delta Y}{\Delta I}$$

(xi) A tax is a compulsory payment to the government by the households, firms or other institutional unit. Tax is a revenue receipt because it does not involve any corresponding liability for the government. Examples of Tax receipts are – Progressive and Regressive tax, Direct and indirect tax, Value added tax or Specific tax, etc.

(xii) The above statement is false regarding undistributed profit. It is a part of profit which is not distributed among the factors of

production. Profit is a component of domestic factor income. That's why it is included in domestic factor income and national income.

(xiii) Under oligopoly, the exact behaviour pattern of a producer cannot be determined with certainty. So, demand curve faced by an oligopolistic is indeterminate or uncertain.

(xiv) Under perfect competition market, firms are price taker and firm is free to sell any quantity at this price. This makes the AR curve perfectly elastic and thus parallel to the x – axis. According to AR and MR relationship, when AR is constant, MR = AR. So AR curve is also the MR curve of the form.

(xv) Small c stands for Autonomous consumption expenditure. What does 0<c<1 signify that autonomous consumption expenditure always greater than zero and it never be negative. It increases with the increase in income of individual.

(xvi) 0<c<1 it signify the autonomous consumption expenditure in consumption function. C stands for autonomous consumption level whether income level is zero.

Section - B [32 Marks]

Answer 2

(i) When production of tomato damaged, it means supply become scarce, their prices increase. In such case, the product can be sold at a price greater than the equilibrium price. This is the fundamental economic theory that when demand exceeds supply, prices tend to rise. There is an inverse relationship between the supply and prices of goods and services when demand is unchanged.

(ii) Given, Initial demand (Q) = 100 units

Rise in price = ₹ 5

Fall in demand = 5 units

New demand = 95 units

Elasticity of Demand = 1.2

Initial Price = ?

According to the formula

$$\text{Elasticity} = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

$$1.2 = 5/5 \times P/100$$

$$P = 1.2 \times 5/5 \times 100$$

$$P = 120$$

Thus, Price before change in demand = ₹ 120

Answer 3

(i)

Direct Tax	Indirect Tax
A direct tax is a tax that a person or organization pays directly to the entity that imposed it.	Indirect tax is a tax imposed by the government on a taxpayer for goods and services rendered.
Examples: - Income tax, real property tax, personal property tax, and taxes on assets, all of which are paid by an individual taxpayer directly to the government.	Examples of indirect taxes are excise tax, VAT, and service tax.
Direct taxes are non-transferable taxes paid by the tax payer to the government	Indirect taxes are transferable taxes where the liability to pay can be shifted to others.

(ii) Redemption of debt refers to the repayment of a public loan. The simplest measure of debt redemption is to impose new taxes and get the required revenue to repay the loan principal as well as the interest. Refunding and Debt conversion are two important methods of Debt redemption.

(a) Refunding of debt implies the issue of new bonds and securities by the government in order to repay the matured loans. Refunding is the process by which the government raises new bonds to pay off the maturing bonds. In the refunding process, usually short-term securities are replaced by issuing long-term securities. Under this method the money burden of public debt is not relinquished but it is accumulated owing to the postponement of debt redemption.

(b) Conversion of loans is another method of redemption of public debt. It means that an old loan is converted into a new loan. Under this system, high-interest public debt is converted into low-interest public debt. Dalton felt that debt conversion actually relaxes the debt burden.

Answer 4

(i) Marginal costs refer to addition to total cost when one more unit of output is produced.

$$MC_n = TC_n - TC_{n-1}$$

Marginal costs are based on cost of hiring purchasing factors of production that are variable, like – labour, materials, and equipment, for example – and not fixed cost the company will have whether it increase production or not. Marginal cost is the change in total cost when one more unit of output is produced. However, when change in unit produced is more than one, then marginal cost can also be calculated as –

$$MC = \frac{\text{Change in Total Cost}}{\text{Change in units of Output}} = \frac{\Delta TC}{\Delta Q}$$

(ii)

Output (Units)	Total Cost (₹)	Total Fixed Cost	Marginal Cost	Average Cost
0	100	100		
1	240	100	140	240
2	360	100	120	180
3	540	100	180	180
4	620	100	80	155

Formula for the above calculation

$$MC_n = TC_n - TC_{n-1}$$

$$AC = \frac{TC}{Q}$$

Note – Total cost of Zero level of output will be the Total fixed cost for entire unit of production in short run.

Answer 5

(i) Perfect competition refers to a market situation where are very large number of buyers and sellers dealing in a homogeneous product at a price fixed by the market. Features of perfect competition market are as follows:

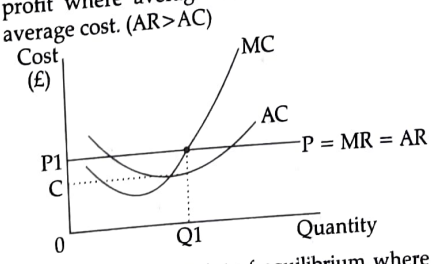
(a) **In a perfect competitive market, there are very large number of buyers and sellers**

(b) **The product offered for sale in the market is homogeneous.**

(ii) Intense competition refers to competition among limited firm or seller. Oligopoly is a market situation in which there are few firms selling homogeneous or differentiated product.

OR

A firm in equilibrium will earn supernormal profit where average revenue is more than average cost. (AR>AC)



(a) Point E is the point of equilibrium where MR = MC and MC > MR after equilibrium.

(b) OQ_1 is the equilibrium level of output as here MR = MC

(c) At this level of output, the firm enjoy super normal profit.

(d) The firm makes super normal profit as here, AR = EQ and AC = AQ_1

(e) Super Normal Profit (per unit) = AR – AC or EQ – AQ = EA

Hence, total super normal profit EA× OQ = $EACP_1$

Answer 6

(i) Yes it is true, in the short run under perfect competition, a firm should produce if AR ≥ AVC because, by covering variable costs, the firm is incurring the loss of fixed cost only which it has to incur even when production is discontinued. But in long run, firm will not continue, if it is incurring losses.

(ii) In the long run, profits and losses are eliminated because an infinite number of firms are producing infinitely divisible, homogeneous products. Free entry will shift the market supply curve to the right driving down the market price and profits or there will be more competition which will lower price and profits. Similarly, free exit of the firm will reduce supply, supply curves shifts left. Due to the above reasons, at equilibrium level, the firms earn only normal profits in long run.

Answer 7

(i) Inflation means the increase in the prices of goods and services in an economy. It refers to a general rise in the level of prices. Types of Inflation:

(a) **Creeping inflation:** Creeping inflation or mild inflation is the situation when the rate of inflation is between 2% to 3%.

(b) **Walking or Trotting inflation:** Walking inflation or Trotting inflation is the situation when the rate of inflation is between 3% to 8%

(c) **Running inflation:** Running inflation is the situation of inflation when the rate of inflation is between 10% to 20%.

(d) **Hyper or Galloping inflation:** Hyper or Galloping Inflation is the situation of inflation when the rate of inflation is between 20% to 30% or even more.

(ii) The rate at which the RBI offers short period loan to the commercial bank by buying the government securities in the open market is called Repo Rate. During inflation, the cost of capital is increased by increasing the repo rate. This lower the demand for credit and accordingly, the supply of money in the economy, as desired.

Answer 8

(i) Commercial bank receive deposits from the public. The depositors are free to withdraw, in part or full , their deposit amounts by writing cheques. The bank use the money in these deposits to give loans. These functions of the commercial banking system are the basis of deposit creation or money creation.

Suppose initially people deposit 100. The bank use this money for giving loan. But the banks cannot use the whole of deposit for this purpose. Suppose initially people deposit 100 and the LRR is 20 percent. Further suppose that banks keep only now free to lend the

remainder 80. Suppose they lend 80. This increase demand deposits in banks by 80. It is 80 percent of the initial deposit. These deposits of 80 have resulted on account of loan given by the bank. Let's see this process in the below table

Round	Deposit	Loan	Cash Reserve (LRR = 0.2)
Initial	100	80	20
	80	64	16
	64	51.20	12.80
Total	500	400	100

In the above illustration, LRR is 0.2, therefore

Money multiplier $= \frac{1}{LRR}$ or $\frac{1}{0.2} = 5$

The total money creation= initial deposit $\frac{1}{LR}$

$$100 \quad \frac{1}{0.2} = 500$$

(ii) Qualitative Methods of credit control refers to the monetary policy by the central bank which includes those instruments that focus on the selected sectors of the economy to control and regulate the money supply in these sectors. RBI implies Qualitative Methods of credit control by following ways –

(a) **Changes in margin requirements:** Changes in margin requirements are designed to influence the flow of credit against specific commodities. The commercial banks generally advance loans to their customers against some security or securities offered by the borrower and acceptable to banks. A rise in the margin requirement results in a contraction in the borrowing value of the security and similarly, a fall in the margin requirement results in expansion in the borrowing value of the security.

(b) **Regulation of Consumer Credit:** Regulation of consumer credit is designed to check the flow of credit for consumer durable goods. This can be done by regulating the total volume of credit that may be extended for purchasing specific durable goods and regulating the number installments through which such loan can be spread. Central Bank uses this method to restrict or liberalize loan conditions accordingly to stabilize the economy.

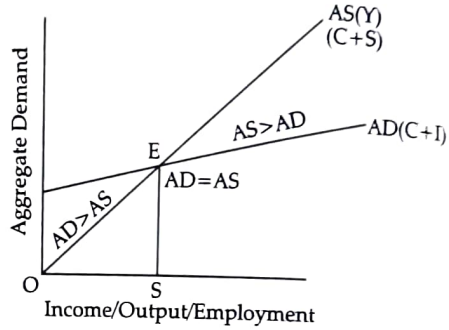
(c) **Credit Rationing:** Rationing of credit is a method by which the Central Bank seeks to limit the maximum amount of loans and advances and, also in certain cases, fix ceiling for specific categories of loans and advances.

(d) **Moral Suasion:** Moral suasion and credit monitoring arrangement are other methods of credit control. The policy of moral suasion will succeed only if the Central Bank is strong enough to influence the commercial banks.

Answer 9

According to Keynes, equilibrium level of income is determined by that point where aggregate demand is equal to aggregate supply. Symbolically, the equilibrium condition is given as AD = AS

C+I = C+S



In the figure,

AD = The aggregate demand curve is the vertical summation of consumption (C) and investment (I) demand schedule at each level of income.

AS = The aggregate supply or national income curve is the sum total of consumption and saving.

E = The point of Equilibrium or point of effective demand, where AD = AS or C+I = C+S

Equilibrium occurs when planned spending equals planned output.

Section - C [32 Marks]

Answer 10

(i) Law of variable Proportion states that as we increase quantity of only one input, variable input, keeping other input fixed, total product initially increases at an increasing rate, then at a decreasing rate and finally at a negative rate. According to the Law of variable Propitiation, the changes in TP and MP can be classified into following three phases:

(a) **Phase – I (Increasing Return to a Factor).** In the first phase, every additional variable factor adds more and more to the total output. It means TP increases at an increasing rate and MP of each variable factor rises.

(b) **Phase – II (Diminishing Return to a Factor).** In the second phase, every additional variable factor adds lesser and lesser amount of output. It means, TP increases at

diminishing rate and MP falls with increase in variable factor.

(c) **Phase – III (Negative Return to a Factor).**

In the third phase the employment of additional variable factor causes TP to decline. MP now become negative.

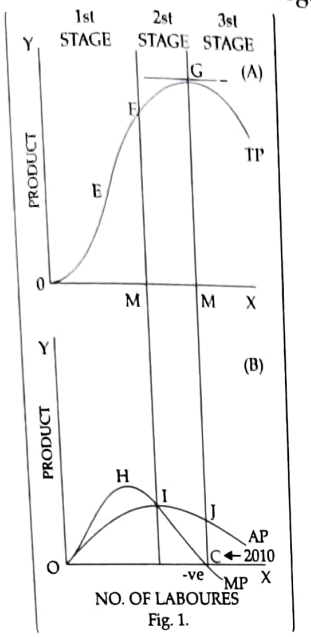


Fig. 1.

(ii) Phase–II (Diminishing Return to a Factor) is very crucial as a rational producer will always aim to produce in this phase because total production is maximum and marginal production of each variable factor is positive. MP falls in this phase, it happens because after a level of output, pressure on fixed input leads to fall in productivity of the variable input.

Answer 11

(i)

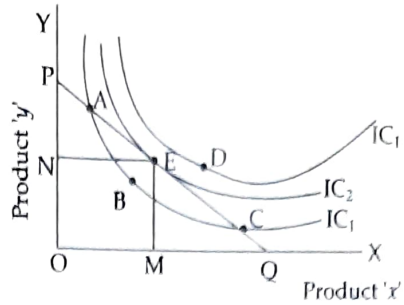
Total Utility	Marginal Utility
Total utility refers to the total satisfaction obtained from all the units of commodity	Marginal utility is the additional utility derived from the consumption additional unit of commodity.
Total utility rises as more consumption is done.	Marginal utility diminishes with an increase in total utility.

(ii) According to Indifference Curve approach, consumer's equilibrium is determined if the following two conditions are satisfied:

(a) MRSxy = Px / Py

(b) MRSxy is declining.

MRS_{xy} is the rate at which the consumer is willing to sacrifice Y to obtain one more unit of X.



Thus, we can say that, "A consumer is in equilibrium at a point where budget line is tangent to Indifference Curve".

Slope of Indifference Curve = Slope of budget line, i.e., $MRS_{xy} = P_x/P_y$

In the diagram, equilibrium is at point E, where the budget line touches the highest attainable indifference curve IC_2 within consumer's budget. Bundles on the Indifference Curve IC_3 are not affordable within budget. Bundles on the Indifference Curve IC_1 (i.e., points A and C) are lying on a lower Indifference Curve i.e. will have lower utility levels as compared to the tangency point E. Therefore, the consumer will choose only the tangency point on the budget line.

Therefore, E is a point of consumer's equilibrium where he maximises his satisfaction. Point E is also called the "Optimum Consumption Point" where he consumes OM of X and ON of Y.

If MRS_{xy} > MRE it implies that the consumer is willing to sacrifice more unit of N than what market requires. This induces the consumer to buy more of M. When he buys more of M, utility derived from M falls and he is willing to sacrifice less of N. Thus, MRS_{xy} starts declining. He continues to consume more of M, till MRS_{xy} = MRE = P_x/PY.

If MRS_{xy} < MRE, it implies consumer is willing to sacrifice less units of Y than what the market requires. He decreases the consumption of M. Due to this MRS_{xy} begins to rise. He continues to decrease the consumption of X till MRS_{xy} = MRE.

Answer 12

(i)

GDPmp	NNPfc
GDPmp is the money value of all final goods and services produced within the domestic territory of a country during an accounting year.	NNPfc is defined as the measure of the factor earnings of the residents of a country, both from economic territory and abroad.
It can be calculated as follows: $GDP^{mp} = \Sigma GVA$ (of all sectors)	It can be calculated as follows: $NNP_{fc} = NDP_{fc} + NFIA$.

(ii) Domestic Income (NDPFC) = Rent + interest + Profit + Wage and salaries + Mixed income of self-employed + Employer's contribution to social security schemes - Consumption of fixed capital

= (1) + (2) + (3) + (4) + (8) + (9) - (5)

= 120 + 15 + 45 + 330 + 360 + 30 - 150

= ₹ 750 crore

National Income (NNP_{FC}) = Domestic Income + Net factor income from abroad

= NDPFC + (Factor income earned abroad - Factor income paid abroad)

= 750 + (30 - 60)

= ₹ 720 crore

OR

(i)

Basic	Private Income	National Income
Meaning	Private income is the income generated by any private individual or a household from engaging in any occupational activities or any type of income that is not received as salary or commission	National income is the total income that is generated by all the economic activities taking place in an economy during a financial year
Meaning Sectors Involved	It includes income that is generated from the private sector only	It includes income that is generated from both public and private sectors
Compo-nents	Consists of transfer earnings along with factor incomes	Consists of only factor incomes
Formula	Private Income = Income from domestic product accruing to private sector + Net factor income from abroad + All types of transfer incomes	National Income = Rent + Compensation + Interest + Profit + Mixed income

(Any two)

(ii) GNPMP = Government final consumption expenditure + Net factor income earned from abroad + (Export - Import) + Private final consumption expenditure + Net domestic capital formation + Consumption of fixed capital.

= 800 + (-) 10 + (250 - 300) + 1200 + 385 + 85

= ₹ 2410 crore

NDPFC = GNPMP - (Export - Import) - Consumption of fixed capital - Net Indirect taxes

= 2410 + 50 - 85 - 70

= ₹ 2305 crore

Answer 13

(i) The balance of payments of a country is a systematic record of all economic transactions between its residents and rest of the world during a given period of time.

(ii) As per the above information, Deficit in the balance of payments may be caused due to number of factors.

(a) The most important factor is Domestic inflation. When there is inflation in the domestic economy, foreign goods become relatively cheaper as compared to domestic goods. It increases imports which causes a deficit in the BOP.

(b) When the domestic economy is going through a phase of boom, then domestic production may be unable to satisfy the domestic demand. It leads to a deficit in BOP, due to increase in imports.

(iii) A flexible exchange-rate system is a monetary system that allows the exchange rate to be determined by supply and demand. It is also called 'free exchange rate' as it is determined by the free play of supply and demand forces in the international money market.

(iv) Foreign Currency is the spine of international investments and global trading. Without it, it would be nearly impossible to determine the value of goods and services imported and exported by different countries to each other. A country, like India need reserves to pay external debts, affords capital to fund sectors of the economy, and profit from diversified portfolios. Countries use foreign currency reserves to keep a fixed rate value, maintain competitively priced exports, remain liquid in case of crisis, and provide confidence for investors.

(v) For the determination of foreign exchange rate under floating regime, the market forces of supply and demand for foreign exchange economic variables are represented along x-axis and y-axis.

■■■